

Procedure Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To create and execute stored procedures in MySQL, using IN, OUT, and INOUT parameters and control flow.

MySQL needs DELIMITER changed before creating a procedure, since the body itself contains semicolons. Change it back to ; afterwards.

-- 1. PROCEDURE: Basic procedure with no parameters

```
DELIMITER //
CREATE PROCEDURE ShowAllFaculty()
BEGIN
    SELECT * FROM FACULTY;
END //
DELIMITER ;

CALL ShowAllFaculty();
```

```
mysql> CALL ShowAllFaculty();
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	Dept	Age
F001	Ramesh	M	1978-05-14	2005-06-01	9840012345	CSE	48
F002	Mohan	M	1965-11-02	1998-07-15	9840012346	CSE	59
F003	Sunitha	F	1980-03-22	2008-08-20	9840012347	ECE	46
F004	Kavya	F	1958-01-10	1990-06-10	9840012348	IT	68
F005	Arjun	M	1975-09-30	2003-09-01	9840012349	CSE	51

```
5 rows in set (0.03 sec)
```

-- 2. PROCEDURE with IN parameter

```
DELIMITER //
CREATE PROCEDURE GetFacultyByDept(IN deptName VARCHAR(10))
BEGIN
    SELECT FacNo, FacName, Age FROM FACULTY WHERE Dept = deptName;
END //
DELIMITER ;

CALL GetFacultyByDept('CSE');
```

```
mysql> CALL GetFacultyByDept('CSE');
+-----+-----+-----+
| FacNo | FacName | Age |
+-----+-----+-----+
| F001  | Ramesh  | 48  |
| F002  | Mohan   | 59  |
| F005  | Arjun   | 51  |
+-----+-----+-----+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

-- 3. PROCEDURE with OUT parameter

```
DELIMITER //
CREATE PROCEDURE CountFaculty(OUT total INT)
BEGIN
    SELECT COUNT(*) INTO total FROM FACULTY;
END //
DELIMITER ;
```

```
CALL CountFaculty(@n);
SELECT @n AS TotalFaculty;
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE CountFaculty(OUT total INT)
-> BEGIN
->     SELECT COUNT(*) INTO total FROM FACULTY;
-> END //
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL CountFaculty(@n);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT @n AS TotalFaculty;
+-----+
| TotalFaculty |
+-----+
|             5 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> |
```

-- 4. PROCEDURE with IN and OUT parameters together

```
DELIMITER //
CREATE PROCEDURE AvgAgeByDept(IN deptName VARCHAR(10), OUT avgAge DECIMAL(5,2))
BEGIN
    SELECT AVG(Age) INTO avgAge FROM FACULTY WHERE Dept = deptName;
END //
DELIMITER ;

CALL AvgAgeByDept('CSE', @avg);
SELECT @avg AS AverageAge;
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE AvgAgeByDept(IN deptName VARCHAR(10), OUT avgAge DECIMAL(5,2))
-> BEGIN
->     SELECT AVG(Age) INTO avgAge FROM FACULTY WHERE Dept = deptName;
-> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL AvgAgeByDept('CSE', @avg);
Query OK, 1 row affected, 1 warning (0.00 sec)

mysql> SELECT @avg AS AverageAge;
+-----+
| AverageAge |
+-----+
|      52.67 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

-- 5. PROCEDURE with INOUT parameter

```
DELIMITER //
CREATE PROCEDURE DoubleValue(INOUT val INT)
BEGIN
    SET val = val * 2;
END //
DELIMITER ;

SET @x = 10;
CALL DoubleValue(@x);
SELECT @x AS DoubledValue;
```

```

mysql> DELIMITER ;
mysql>
mysql> SET @x = 10;
Query OK, 0 rows affected (0.00 sec)

mysql> CALL DoubleValue(@x);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @x AS DoubledValue;
+-----+
| DoubledValue |
+-----+
|           20 |
+-----+
1 row in set (0.00 sec)

```

-- 6. PROCEDURE with conditional logic (IF / ELSEIF)

```

DELIMITER //
CREATE PROCEDURE CheckDeptSize(IN deptName VARCHAR(10))
BEGIN
    DECLARE cnt INT;
    SELECT COUNT(*) INTO cnt FROM FACULTY WHERE Dept = deptName;
    IF cnt > 2 THEN
        SELECT 'Large department' AS Result;
    ELSEIF cnt > 0 THEN
        SELECT 'Small department' AS Result;
    ELSE
        SELECT 'No such department' AS Result;
    END IF;
END //
DELIMITER ;

CALL CheckDeptSize('CSE');

```

```

mysql> DELIMITER //
mysql> CREATE PROCEDURE CheckDeptSize(IN deptName VARCHAR(10))
    -> BEGIN
    ->     DECLARE cnt INT;
    ->     SELECT COUNT(*) INTO cnt FROM FACULTY WHERE Dept = deptName;
    ->     IF cnt > 2 THEN
    ->         SELECT 'Large department' AS Result;
    ->     ELSEIF cnt > 0 THEN
    ->         SELECT 'Small department' AS Result;
    ->     ELSE
    ->         SELECT 'No such department' AS Result;
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL CheckDeptSize('CSE');
+-----+
| Result          |
+-----+
| Large department |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```

-- 7. PROCEDURE with a loop (WHILE)

```

DELIMITER //
CREATE PROCEDURE PrintNumbers(IN upto INT)
BEGIN
    DECLARE i INT DEFAULT 1;
    WHILE i <= upto DO
        SELECT i;
        SET i = i + 1;
    END WHILE;
END //
DELIMITER ;

CALL PrintNumbers(3);

```

```

mysql> DELIMITER //
mysql> CREATE PROCEDURE PrintNumbers(IN upto INT)
    -> BEGIN
    ->     DECLARE i INT DEFAULT 1;
    ->     WHILE i <= upto DO
    ->         SELECT i;
    ->         SET i = i + 1;
    ->     END WHILE;
    -> END //
ERROR 1304 (42000): PROCEDURE PrintNumbers already exists
mysql> DELIMITER ;
mysql>
mysql> CALL PrintNumbers(3);
+-----+
| i      |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      2 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```

-- 8. Viewing and dropping procedures

```
SHOW PROCEDURE STATUS WHERE Db = 'lab';
SHOW CREATE PROCEDURE ShowAllFaculty;
DROP PROCEDURE IF EXISTS ShowAllFaculty;
```

```
mysql> SHOW PROCEDURE STATUS WHERE Db = 'lab';
```

Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation_connection
lab	AvgAgeByDept	PROCEDURE	root@localhost	2026-08-08 14:06:34	2026-08-08 14:06:34	DEFINER		cp850	cp850
lab	CheckDeptSize	PROCEDURE	root@localhost	2026-08-08 14:07:36	2026-08-08 14:07:36	DEFINER		cp850	cp850
lab	CountFaculty	PROCEDURE	root@localhost	2026-08-08 14:06:05	2026-08-08 14:06:05	DEFINER		cp850	cp850
lab	DoubleValue	PROCEDURE	root@localhost	2026-08-08 14:06:56	2026-08-08 14:06:56	DEFINER		cp850	cp850
lab	GetFacultyByDept	PROCEDURE	root@localhost	2026-08-08 14:04:56	2026-08-08 14:04:56	DEFINER		cp850	cp850
lab	PrintNumbers	PROCEDURE	root@localhost	2026-08-08 14:08:17	2026-08-08 14:08:17	DEFINER		cp850	cp850
lab	ShowAllFaculty	PROCEDURE	root@localhost	2026-08-08 14:02:32	2026-08-08 14:02:32	DEFINER		cp850	cp850

7 rows in set (0.03 sec)

```
mysql> SHOW CREATE PROCEDURE ShowAllFaculty;
```

Procedure	sql_mode	Create Procedure
ShowAllFaculty	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION	CREATE DEFINER= BEGIN SELECT * FROM FACULTY; END

1 row in set (0.02 sec)

```
mysql> DROP PROCEDURE IF EXISTS ShowAllFaculty;  
Query OK, 0 rows affected (0.01 sec)
```